UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PRE CALCULUS
COURSE CODE	:	MAT133
EXAMINATION	:	OCTOBER 2026
TIME	:	2 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (October 2026).
   Computed answers show working; every answer is produced by the deterministic MAT133 generator and verified by code where possible. No model answer is presented as official.

2. Every marking point comes from the generator's worked solution; M1/A1 codes are aligned from the final answer backwards.

3. Five questions; total = 67 marks.

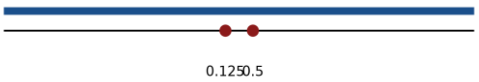
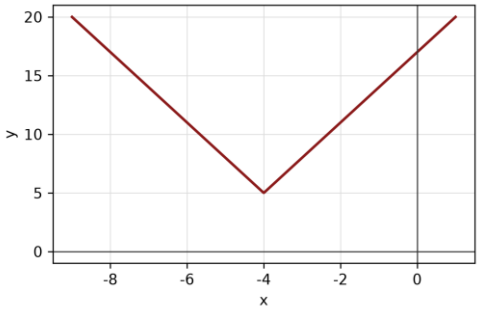
4. Computed: 15 part(s). Placeholders: 0.
   Questions and answers are for local study use only.

5. Generated deterministically from the 41 MAT133 past papers; seed 2.

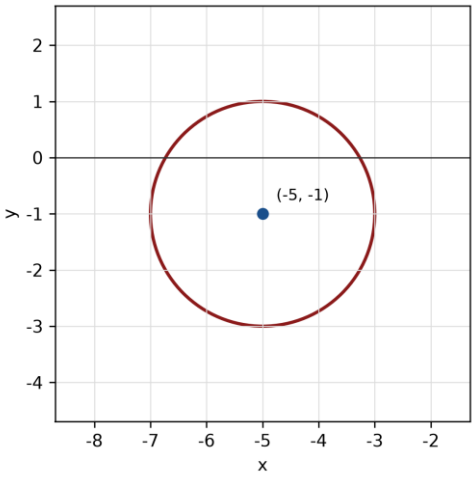
FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

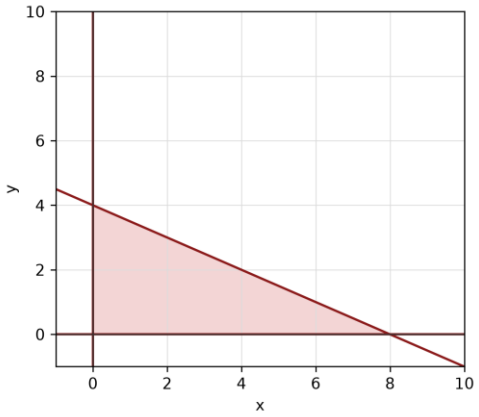
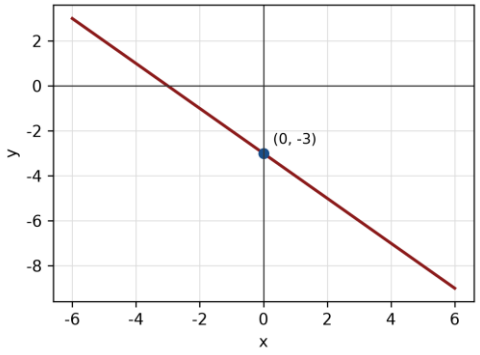
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NO	SOLUTION		MARK
1 a)	$3 + \frac{-2x-2}{1-2x} \leq 0$	M1 A1	5 m
	$\frac{1-8x}{1-2x} \leq 0$	A1	
	$\left[\frac{1}{8}, \frac{1}{2}\right)$	A1	
			
1 b)	$\lvert 3x+5 \rvert - 6 > 0$	M1	5 m
	$3x+5 > 6$ or $3x+5 \leq -6$	A1	
	$\left(-\infty, -\frac{11}{3}\right) \cup \left(\frac{1}{3}, \infty\right)$	A1	
			
1 c)	$y = 3\lvert x+4 \rvert + 5$	B1	5 m
	vertex $(-4, 5)$	A1	
	no real intercepts	A1	
			
2 a)	$(f+g)(x) = 7x+1$	M1	4 m
	$(f-g)(x) = 3x+7$	A1	
	$(fg)(x) = 10x^2 - 7x - 12$	B1	

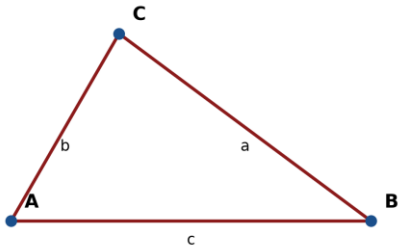
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NO	SOLUTION		MARK
2 b)	$4x + -1 \neq 0$	M1	4 m
	$x \neq \frac{1}{4}$	A1	
	$2x + -4 \geq 0$	M1	
	$x \geq 2$	A1	
2 c)	$x^2 + 10x + y^2 + 2y = -22$	M1 M1	5 m
	$(x+5)^2 + (y+1)^2 - 26 = -22$	A1	
	$(x+5)^2 + (y+1)^2 = 4$	A1	
			
3 a)	$(-4+i)\times(-2+6i)$	M1 A1	4 m
	$2-26i$	A1	
3 b)	$i^{19} = i^{4(4)+3}$	M1	3 m
	$= (i^4)^4 i^3 = 1^4 i^3$	A1	
	$= -i$	A1	
3 c)	$x^3 - 2x^2 + 6x - 4 = (x-2)(x^2+6) + 8$	M1 M1 A1	5 m
	$q(x) = x^2 + 6, \quad r = 8$	A1	

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NO	SOLUTION		MARK
4 a)	$4 = x^2 - 5$	M1	5 m
	$x^2 - 9 = 0$	A1	
	$(x-3)(x+3) = 0$		
	$x = -3$ or $x = 3$		
	$(-3, 4), \quad (3, 4)$	A1	
4 b)	$x = 0, \quad y = 0, \quad 1x + 2y = 8$	B1 B1	5 m
	$x \geq 0, \quad y \geq 0, \quad 1x + 2y \leq 8$	A1	
			
4 c)	$-4x + 4y - 5 = 0$	M1	5 m
	$y = x + \frac{5}{4}$	M1	
	$m_{\text{new}} = -1$		
	$y = -x - 3$	A1	
			

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NO	SOLUTION		MARK
5 a)	$\cos^2(x) - \frac{3\cos(x)}{2} + \frac{1}{2} = 0$	M1	5 m
	$(\cos(x) - 1)\left(\cos(x) - \frac{1}{2}\right) = 0$	A1	
	$\cos x = \frac{1}{2}$ or $\cos x = 1$		
	$x = 0^\circ, 60^\circ, 300^\circ$	A1	
5 b)	$30^\circ \times \frac{\pi}{180} = \frac{\pi}{6}$	M1 A1	3 m
	$\frac{\pi}{6} \times \frac{180}{\pi} = 30^\circ$	A1	
5 c)	$a^2 = b^2 + c^2 - 2bc\cos A$	M1	4 m
	$a^2 = 6^2 + 10^2 - 2(6)(10)\cos 60^\circ = 76$	A1	
	$a = 2\sqrt{19}$	A1	
			

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